

Malek Hassan

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EDUCATION

CODE University of Applied Sciences <i>B.Sc. Software Engineering - final year</i> Transferred from RWTH Aachen University to complete the final year of the degree in Berlin, moving to a project-based curriculum built around shipping real software.	2026 - Present <i>Berlin, Germany</i>
RWTH Aachen University <i>B.Sc. Computer Science - first two years</i>	Oct 2023 - 2026 <i>Aachen, Germany</i>

EXPERIENCE

African Arab International Bank <i>Software Engineering Intern, Digital Factory</i> - Built an end-to-end banking support assistant with HTML, CSS, JavaScript/TypeScript, URL-scraped public knowledge, and locally hosted Ollama inference. - Won adoption of the concept by the development team and documented its retrieval, inference, and data flow for the bank's internal review process.	Jul 2025 - Sep 2025 <i>Cairo, Egypt</i>
RWTH Aachen University <i>Student Technical Support, CS Peer Assistance</i> Supported peers with Java debugging, Git/GitHub workflows, and LaTeX document structure.	Oct 2023 - Jan 2024 <i>Aachen, Germany</i>

SELECTED PROJECTS

MOZCODE - Symbol-Level Context Engine for Coding Agents <i>Creator and maintainer MIT licensed</i> - Built a local MCP server for Claude Code, Codex, and OpenAI models that uses tree-sitter ASTs to return requested symbols and structural outlines instead of entire source files. - Implemented grouped code search, AST-anchored edits with parse validation, graceful fallbacks, and cached read-only SQLite/PostgreSQL schema introspection. - Instrumented an append-only JSONL ledger and self-contained SVG dashboard; measured an estimated 1,749,436 input tokens avoided across 938 optimized calls, with 80% average payload reduction versus disclosed whole-file/plain-grep baselines.	2026 <i>github.com/Malek1414/MOZCODE</i>
Second Brain - End-to-End RAG Knowledge System <i>Personal knowledge capture and cited retrieval</i> - Built one retrieval loop from voice/text capture through embedding and ChromaDB indexing to structured Markdown synchronization with an Obsidian vault. - Implemented an Animus chat interface that retrieves passages from the user's corpus and generates Claude answers with citations back to the source notes. - Integrated Node.js, Express, LangChain/LlamaIndex, ChromaDB, the Anthropic Claude API, Obsidian vault sync, and a vanilla JavaScript/Canvas interface.	2025 <i>Node.js ChromaDB Claude API</i>

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C, HTML/CSS
AI and systems: RAG, Model Context Protocol, tree-sitter, vector search, prompt engineering, Claude API, Ollama
Backend and data: Node.js, Express, PostgreSQL, SQLite, Supabase, ChromaDB, REST APIs
Frontend and visualization: React, Next.js, Vite, D3.js, Recharts, Tailwind CSS, Canvas, Framer Motion
Developer tooling: Git/GitHub, Vitest, esbuild, PDF processing, command-line workflows

CERTIFICATIONS

Anthropic: Introduction to Model Context Protocol; Model Context Protocol - Advanced Topics; Introduction to Agent Skills; Introduction to Subagents (May 2026)

LEADERSHIP

Organising Committee Head, Model United Nations <i>High school leadership role</i> - Led the organising committee across sales, marketing, logistics, and administration, and delivered a keynote to an audience of 300+. - Improved the event's operational performance by 20% year over year.	Sep 2022 - Jun 2023 <i>Cairo, Egypt</i>
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AWARDS AND HONORS

State Champion, Basketball - won a national/state championship title in the youth division.

MOZCODE figures are estimates against local counterfactual baselines, not billing data. Snapshot: 6 Aug 2026.